

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Explain the structure of Mobile Computing Application.	CO2	6
B) Explain the characteristics of Mobile Computing.	CO1	6
C) What is Mobile Computing? Explain its applications in the real-world scenario	CO3	6
Q.2 Solve Any Two of the following.		12
A) Describe in detail about TDMA, FDMA, and CDMA and tabulate the differences among them.	CO4	6
B) Discuss the various Reservation Based schemes.	CO1	6
C) Summarize the issues in the Wireless MAC Protocols.	CO5	6
Q.3 Solve Any Two of the following.		12
A) Write short note on GSM.	CO2	6
B) What are the features of mobile IP. Explain in brief.	CO4	6
C) Explain Overview of Mobile IP in detail.	CO5	6
Q.4 Solve Any Two of the following.		12
A) What is ad hoc network? Explain its applications.	CO1	6
B) Write short note on Universal Mobile Telecommunication System (UMTS).	CO3	6
C) What is VANET. Explain in detail.	CO4	6
Q.5 Solve Any Two of the following.		12
A) Explain Mobile Device Operating Systems & its special constrains in detail.	CO1	6
B) Explain MANET vs. VANET	CO5	6
C) Explain the mobile payment system and security issues	CO2	6

*** End ***